

# Product Requirements Document

## Spotify Premium Conversion & Retention Initiative

| Field             | Value                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------|
| Document owner    | Product Manager, Growth & Monetization (Case Study)                                                   |
| Status            | Draft — for review                                                                                    |
| Version           | 1.0                                                                                                   |
| Last updated      | [Update on each revision]                                                                             |
| Stakeholders      | Growth PM (owner), Data Science, Lifecycle Marketing, Ads Product, Engineering, Design, Legal/Privacy |
| Related artifacts | spotify_premium_case_study.ipynb (source analysis notebook)                                           |

### 1. Executive Summary

This PRD proposes a phased initiative to increase Spotify Premium conversion and reduce silent user disengagement, based on an analysis of 50,000 simulated user behavior records (see the accompanying analysis notebook). The analysis identifies a sizable, identifiable segment of engaged-but-unconverted free users, shows that ad interaction correlates with conversion behavior, and demonstrates that inactivity risk can be predicted from existing usage signals. This document translates those findings into four scoped epics, success metrics, an experimentation plan, and a rollout timeline.

### 2. Problem Statement

Subscription revenue is driven by two levers: converting free users to Premium, and retaining the users we already have. Today, neither lever is being pulled as precisely as it could be:

- Under-monetized engaged free users. A subset of free users interact with in-product ads, a sign of active, attentive usage but never convert to Premium. Today there is no systematic way to identify or target this group.
- Silent disengagement. Users who stop listening for 3+ months rarely convert or renew, but there is no early-warning system that flags them while an intervention could still work.
- One-size-fits-all treatment. Free users are treated as a single, homogeneous group in product and marketing, despite showing behaviorally distinct patterns (see Section 5, Personas).

Left unaddressed, these gaps mean the company is paying acquisition and infrastructure costs for free users who are being neither converted nor re-engaged effectively.

### 3. Objectives

1. Increase the Premium conversion rate among identifiably high-intent free users.
2. Reduce the rate of unaddressed 3+ month inactivity by surfacing at-risk users before they go dark.
3. Establish a repeatable, data-driven segmentation so product, marketing, and lifecycle teams target users differently based on behavior, not a single blended average.
4. Ship the above as a phased, testable program — not a single big-bang launch — so each lever's impact can be isolated and measured.

## 4. Goals & Success Metrics

Baseline values for each metric below are produced by Section 10 of the analysis notebook and must be filled in / refreshed at the start of each planning cycle:

| Metric                               | Definition                                                                                               | Source              | Target                                                                                                  |
|--------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------|
| Premium conversion rate              | % of all users on a Premium subscription                                                                 | Notebook §6.1 / §10 | Increase by an agreed-upon relative lift (e.g., +X pp) within 2 quarters of full rollout                |
| High-value free user conversion rate | % of the high-value free segment (engaged, unconverted) that converts to Premium after campaign exposure | Notebook §7.3       | Meaningfully higher than the baseline free-user conversion rate                                         |
| Ad-interaction → conversion rate     | % of ad-interacting users who convert to paid                                                            | Notebook §7.1       | Improve via ad experience changes (Epic 2)                                                              |
| 3+ month inactivity rate             | % of users with 3+ consecutive months of inactivity                                                      | Notebook §6.4 / §10 | Reduce among users flagged "at risk" by the churn model (Epic 3) relative to an unflagged control group |
| Risk model precision/recall          | Precision & recall of the inactivity risk classifier on held-out data                                    | Notebook §9         | Meets a minimum bar agreed with Data Science before triggering any user-facing action                   |

*Note: because the dataset used for this analysis is simulated, absolute target values (e.g., "+3pp") should be set once this analysis is repeated on real production data, not hard-coded from the simulated baseline.*

## 5. Target Users & Personas

Personas below are derived from the K-Means segmentation in Notebook §8 and the high-value free user definition in §7.3. Cluster names and exact profiles should be finalized against the live `segment\_profile` output before this section is considered final.

| Persona                               | Defining behavior                                | Product treatment                                                 |
|---------------------------------------|--------------------------------------------------|-------------------------------------------------------------------|
| High-Value Free User (primary target) | Free tier; interacts with ads; has not converted | Targeted upsell messaging, trial offers (Epic 1)                  |
| Engaged Premium                       | Premium tier; low inactivity; regular usage      | Retention/loyalty, avoid over-messaging                           |
| At-Risk / Dormant                     | Rising months_inactive; flagged by churn model   | Win-back campaigns, re-engagement nudges (Epic 3)                 |
| New / Low-Signal                      | Limited behavioral history to date               | Onboarding-focused messaging; re-evaluate after more data accrues |

## 6. Scope

### 6.1 In Scope

- Definition and productionization of the "high-value free user" segment as a queryable, always-on audience.
- A/B-tested upsell messaging and offers targeted at that segment (in-app + email/push, depending on existing lifecycle tooling).
- Ad experience changes aimed at improving ad-interaction → conversion rate (e.g., contextual upgrade prompts shown after ad interactions).
- A first-pass churn/inactivity risk score, exposed to lifecycle marketing tooling for win-back triggers.
- Segment definitions and naming for the four behavioral clusters, handed off to marketing/product for differentiated treatment.

### 6.2 Out of Scope (this phase)

- Pricing changes or new subscription tiers.
- Real-time (sub-daily) risk scoring — batch/daily scoring is sufficient for v1.
- Any model or segment that hasn't been validated against real (non-simulated) user data.
- Country-specific pricing/marketing experiments — flagged as a candidate for a follow-on PRD pending the geographic analysis in Notebook §6.3.

## 7. Epics & Requirements

### 7.1 Epic 1: High-Value Free User Upsell Campaign (Priority: P0)

Targets free users who interact with ads but have not converted (Notebook §7.3) — the highest-intent, lowest-effort segment to convert, since they are already demonstrably engaged with the product.

#### User stories

- As a free user who regularly engages with the app (including ads), I want to be shown a relevant Premium offer, so that I understand what I'm missing without feeling spammed.
- As a Growth PM, I want the high-value free segment refreshed automatically, so that campaign targeting stays current without manual re-pulls.
- As a marketing lifecycle owner, I want to A/B test offer variants against this segment, so that we learn which incentive structure converts best.

#### Functional requirements

5. Define and materialize the "high-value free user" segment (`premium_user = 0`, `ad_interaction = 1`, `ad_conversion_to_subscription = 0`) as a recurring, queryable audience.
6. Build at least two upsell offer variants (e.g., discounted trial vs. feature-preview messaging) for A/B testing.
7. Instrument exposure and conversion tracking so campaign lift can be measured against a held-out control group.
8. Cap messaging frequency per user to avoid fatigue (exact cap to be set with Lifecycle Marketing).

#### Non-functional requirements

- Segment refresh cadence: daily or weekly, agreed with Data Engineering.
- Respect existing user communication/opt-out preferences and applicable privacy regulations.

### 7.2 Epic 2: Ad Experience & Conversion Optimization (Priority: P1)

Uses the observed relationship between ad interaction and conversion (Notebook §7.1) to test whether ad placements can themselves become a conversion touchpoint, rather than a pure monetization mechanism for free users.

### User stories

- As a free user who just interacted with an ad, I want an easy, low-friction path to try Premium, so that I can immediately remove ads if I want to.
- As an Ads Product Manager, I want to measure whether contextual upgrade prompts cannibalize ad revenue, so that we don't trade one revenue stream for a smaller gain in another.

### Functional requirements

9. Add a contextual "Remove ads with Premium" prompt shown after specific ad interaction events.
10. A/B test prompt presence/placement against a control with no additional prompt.
11. Track both ad revenue and Premium conversion for treatment vs. control to compute net revenue impact, not conversion alone.

### Non-functional requirements

- Any change to ad load or placement must be reviewed with the Ads Product team before rollout.
- Prompt must comply with existing ad-experience and accessibility guidelines.

## 7.3 Epic 3: Churn Early-Warning & Retention System (Priority: P0)

Operationalizes the inactivity risk model (Notebook §9) as a recurring scoring job that flags users trending toward 3+ months of inactivity, so lifecycle marketing can intervene before they go fully dormant.

### User stories

- As a Lifecycle Marketing owner, I want a daily/weekly list of at-risk users ranked by risk score, so that I can trigger win-back campaigns proactively.
- As a Data Scientist, I want the model's precision/recall tracked over time, so that we catch model drift before it degrades campaign targeting.

### Functional requirements

12. Productionize the logistic regression (or an improved successor model) as a scheduled batch scoring job.
13. Expose risk scores to lifecycle marketing tooling via a defined data contract (e.g., a daily table/export).
14. Set a minimum precision/recall bar, agreed with Data Science, below which the model is not used to trigger user-facing actions.
15. Validate the `inactive\_3\_months\_flag` proxy against real subscription-cancellation data as soon as it's available, and re-calibrate the model accordingly.

### Non-functional requirements

- Batch scoring cadence: daily or weekly (real-time explicitly out of scope for v1 — see Section 6.2).
- Model outputs must be auditable (feature importances, not just a black-box score) to support marketing decision-making, per Notebook §9's coefficient analysis.

## 7.4 Epic 4: Segment-Based Personalization (Priority: P2)

Uses the behavioral clusters from Notebook §8 to move product and marketing messaging away from one-size-fits-all treatment of the free/Premium split.

### User stories

- As a Product Marketing Manager, I want named, well-understood user segments, so that I can tailor messaging without re-deriving them myself.
- As a PM on an adjacent team, I want to reuse these segment definitions, so that we don't end up with conflicting user taxonomies across teams.

## Functional requirements

16. Finalize human-readable names and definitions for each of the four clusters, validated against `segment\_profile` (Notebook §8).
17. Publish segment membership as a shared, queryable field alongside `user\_type` and the high-value free flag.
18. Define at least one differentiated treatment per segment (e.g., messaging, offer, or onboarding flow) as a starting point for future experiments.

## Non-functional requirements

- Segment definitions should be re-validated (re-clustered) on a recurring basis, since behavior drifts over time.

## 8. Experimentation Plan

All user-facing changes in Epics 1–3 ship as controlled experiments before any full rollout:

| Epic                          | Experiment design                                                                                                 | Primary metric                                    | Guardrail metric(s)                                        |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------|
| 1. High-Value Free Upsell     | Randomized holdout within the high-value free segment; treatment receives upsell messaging, control receives none | Premium conversion rate (treatment vs. control)   | Unsubscribe/opt-out rate, app-uninstall rate               |
| 2. Ad Experience Optimization | Randomized exposure to contextual upgrade prompt vs. standard ad experience                                       | Net revenue (ad revenue + subscription revenue)   | Ad revenue alone, user-reported ad annoyance if measurable |
| 3. Churn Early-Warning        | Randomized holdout among users flagged "at risk"; treatment receives win-back campaign, control receives none     | 3-month reactivation rate (treatment vs. control) | Messaging fatigue / opt-out rate                           |

*Each experiment should run long enough to reach the minimum sample size for statistical significance given the expected effect size and baseline conversion rate — to be calculated by Data Science before launch, not assumed.*

## 9. Phased Rollout & Timeline

| Phase                | Timeframe  | Scope                                                                                                                                     |
|----------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 0 — Validation | Weeks 1–3  | Re-run analysis on real (non-simulated) data; validate churn proxy against actual cancellation data if available; finalize segment names. |
| Phase 1 — Foundation | Weeks 4–7  | Productionize the high-value free segment and the churn risk scoring job (Epics 1 & 3 data plumbing); no user-facing changes yet.         |
| Phase 2 — Experiment | Weeks 8–12 | Launch A/B tests for Epic 1 (upsell campaign) and Epic 3 (win-back campaign) against held-out control groups.                             |

| Phase                 | Timeframe   | Scope                                                                                                       |
|-----------------------|-------------|-------------------------------------------------------------------------------------------------------------|
| Phase 3 — Optimize    | Weeks 13–16 | Launch Epic 2 (ad experience) experiment; analyze Phase 2 results and decide on full rollout vs. iteration. |
| Phase 4 — Personalize | Weeks 17+   | Roll out Epic 4 segment-based treatments informed by learnings from Phases 2–3.                             |

*Timeframes are illustrative starting points for planning and should be re-baselined against actual team capacity and Data/Engineering dependencies.*

## 10. Risks & Assumptions

| Risk / Assumption                                                                        | Impact                                                                       | Mitigation                                                                                                                            |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Analysis is based on a simulated dataset, not real production data.                      | Findings and target metrics may not transfer to real users.                  | Treat all figures as directional until re-validated on production data in Phase 0.                                                    |
| `inactive_3_months_flag` is a proxy for churn, not an actual cancellation event.         | Risk model may misidentify who is truly at risk of churning vs. subscribing. | Validate against real cancellation data before using the model to trigger high-cost interventions; keep humans in the loop initially. |
| Increased messaging to the high-value free segment could increase opt-outs/unsubscribes. | Net negative on user experience and future addressability.                   | Set frequency caps and monitor opt-out rate as a guardrail metric (Section 8).                                                        |
| Ad experience changes (Epic 2) could reduce ad revenue even if conversion improves.      | Net revenue could be flat or negative.                                       | Measure and gate rollout on net revenue, not conversion rate alone.                                                                   |
| User segments (K-Means clusters) may not remain stable over time.                        | Personalization built on stale segments could mis-target users.              | Re-cluster on a recurring cadence and monitor segment stability.                                                                      |

## 11. Dependencies

- Data Engineering: pipeline to materialize and refresh the high-value free segment and churn risk scores in production.
- Data Science: validation of the churn proxy, model monitoring, and minimum precision/recall sign-off.
- Lifecycle Marketing: campaign tooling and messaging/frequency-cap infrastructure for Epics 1 and 3.
- Ads Product: review and sign-off on any change to ad placements or prompts (Epic 2).
- Legal/Privacy: review of any new user-facing messaging triggered by behavioral scoring.

## 12. Appendix

### 12.1 Data source

Spotify User Behavior and Pattern dataset (Kaggle: sahilislam007/spotify-user-behavior-and-pattern), 50,000 simulated user records.

### 12.2 Related artifacts

- Spotify\_Premium\_Conversion\_Churn\_Case\_Study.ipynb — full analysis notebook supporting every metric and claim in this document.

## 12.3 Glossary

| Term                          | Definition                                                                                                        |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Premium user                  | A user whose `subscription_type` contains "premium".                                                              |
| High-value free user          | A free user who has interacted with an ad but has not converted to a paid subscription.                           |
| Inactivity rate (churn proxy) | % of users flagged `inactive_3_months_flag`, i.e. 3+ consecutive months without listening activity.               |
| User segment                  | One of four behavioral clusters produced by K-Means over age, inactivity, and ad-interaction/conversion features. |